MATLAB Simulink: Three-Phase Induction Motor Analysis

Key Objectives

• Interpret and extract key electrical parameters from a three-phase induction motor nameplate.
• Apply scientific theory and calculations to determine key motor performance values such as torque, efficiency, and slip.
• Model and simulate a 1.1kW 380V 50Hz double squirrel cage motor in MATLAB Simulink.
•Compare theoretical results with simulated outputs to verify performance.
• Interpret Simulink-generated graphs to assess accuracy and identify sources of deviation.

Technical Overview

The project required students to extract motor parameters from a nameplate and use them to calculate operational metrics. Key quantities such as input power, torque, airgap power, and efficiency were derived analytically. These values were used to construct and simulate a custom double squirrel cage induction motor model in MATLAB Simulink. The simulation incorporated scopes and displays for voltage, current, torque, speed, and power measurements. The results were then compared against the calculations to validate performance and model accuracy.

Visual Aids

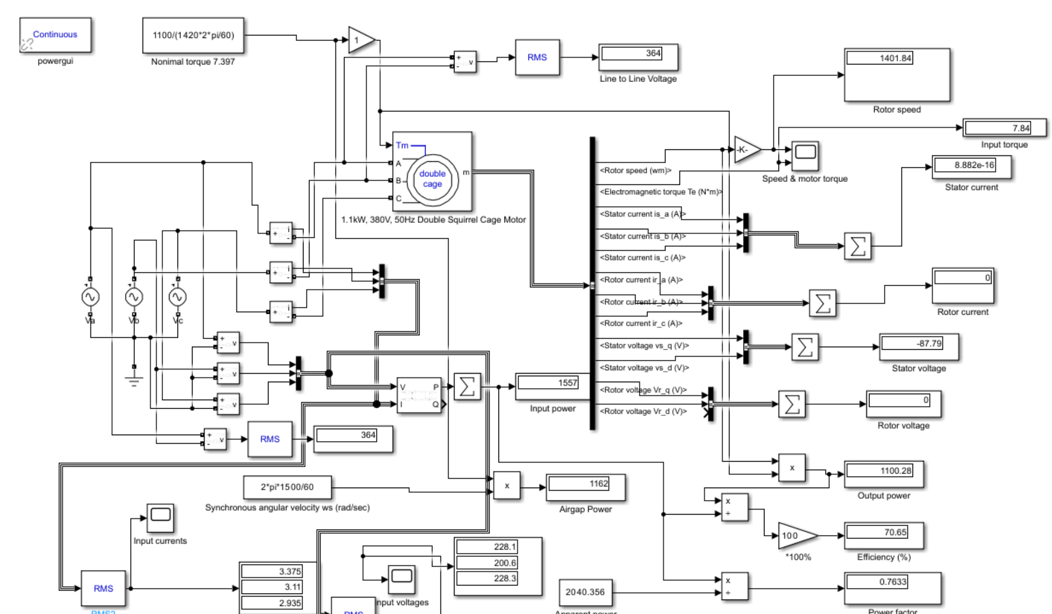

Figure 1: MATLAB Simulink simulation of a Double Squirrel Cage 1.1kW 380V 50Hz motor

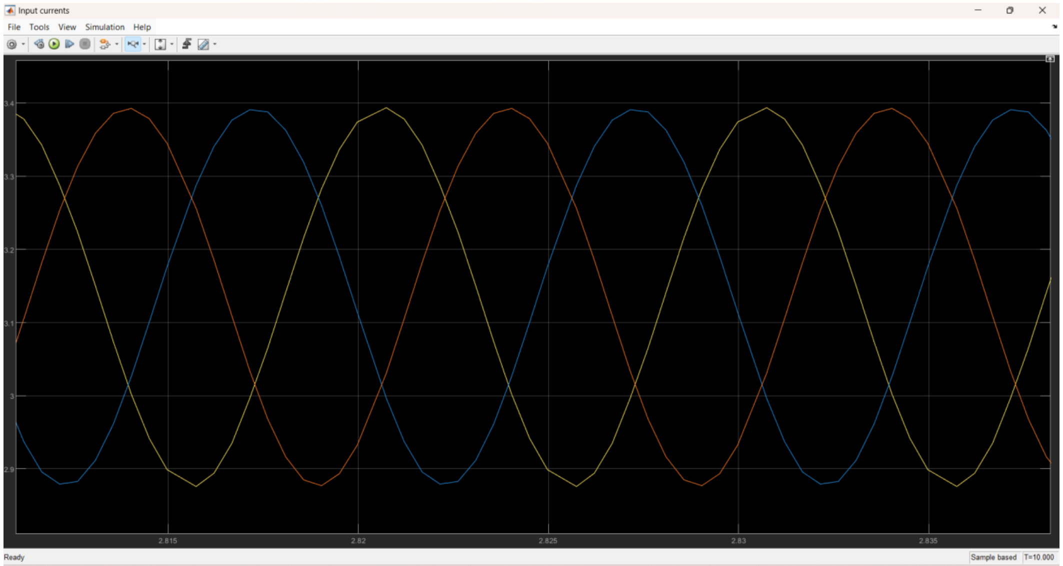

Figure 2: Scope of input currents

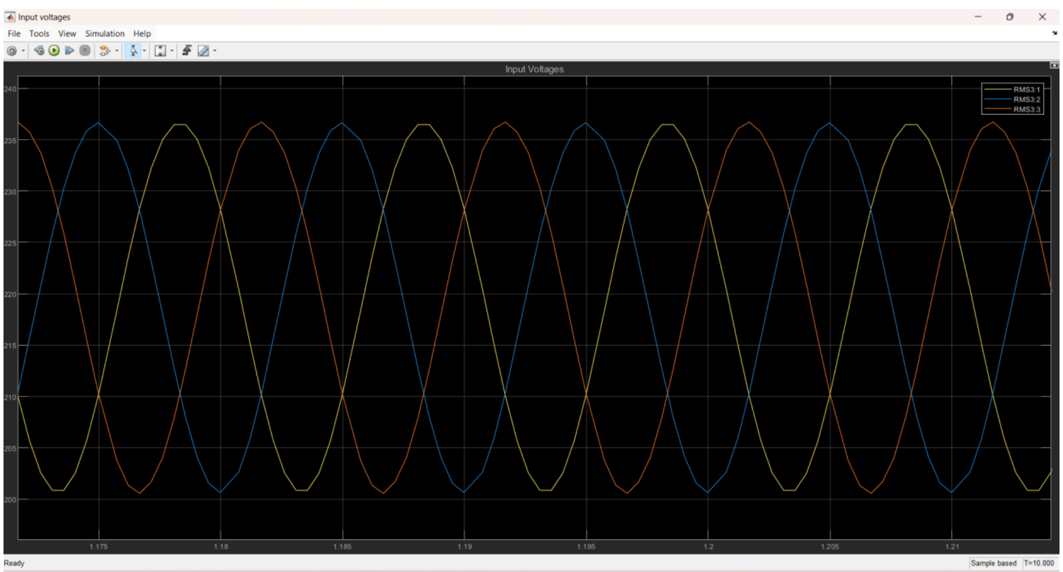

Figure 3: Scope of input voltages

Parameter	Calculated value	Simulated Value
Input Voltage	380V	237V
Input Current	3.1A	3.375A
Input Power	1550.67W	1557W
Output power	1.1 kW	1100.28W
Airgap Power	1267.606 W	1162W
Input torque	9.872 Nm	7.84Nm
Output torque	7.525 Nm	7.397Nm
Synchronous speed	1500 RPM	1500 Nm
Rotor speed	1420 RPM	1401.84RPM
Power factor	0.76	0.7633
Efficiency	70.937%	70.65%

Table 1: Comparison of results

Highlights

• Full motor analysis based on 1.1kW, 380V, 50Hz three-phase AC motor specifications.
• Analytical and simulation methods showed a match within 5% error for most parameters.
• MATLAB Simulink used to build a custom model with scopes for real-time signal monitoring.
• Insights on discrepancies between theoretical and simulated values highlighted potential adjustments in resistance/inductance modeling.

Conclusion

The MATLAB Simulink motor simulation project provided an in-depth comparison between theory and practice. Most calculated parameters aligned closely with simulated results, confirming the model's accuracy. Notable deviations in values like airgap power and input torque were attributed to parameter estimations and modeling assumptions. Overall, the exercise reinforced the value of simulation tools in validating theoretical engineering analyses.